

1.6: Roots of Polynomials

1. Vieta's Formulae (Roots-Coefficient Relations)

Quadratic $ax^2 + bx + c = 0$, roots α, β

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Cubic $ax^3 + bx^2 + cx + d = 0$, roots α, β, γ

$$\sum \alpha = -\frac{b}{a}, \quad \sum \alpha\beta = \frac{c}{a}, \quad \alpha\beta\gamma = -\frac{d}{a}$$

Quartic $ax^4 + bx^3 + cx^2 + dx + e = 0$, roots $\alpha, \beta, \gamma, \delta$

$$\sum \alpha = -\frac{b}{a}, \quad \sum \alpha\beta = \frac{c}{a}, \quad \sum \alpha\beta\gamma = -\frac{d}{a}, \quad \alpha\beta\gamma\delta = \frac{e}{a}$$

The sign pattern is $-, +, -, +, \dots$. $\sum \alpha, \sum \alpha\beta$ denote the elementary symmetric polynomials.

2. Symmetric Function Identities

For quadratic roots:

- $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
- $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$
- $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

- $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

For cubic roots:

- $\sum \alpha^2 = (\sum \alpha)^2 - 2 \sum \alpha\beta$
- $\sum \alpha^3 = (\sum \alpha)^3 - 3(\sum \alpha)(\sum \alpha\beta) + 3\alpha\beta\gamma$

Given $\alpha + \beta = 4$, $\alpha\beta = 3$, find $\alpha^2 + \beta^2$.
 $= (4)^2 - 2(3) = 16 - 6 = 10$.

3. Transforming the Roots

To form a polynomial whose roots are transformed versions of the original, use substitution.

Shift: roots $\alpha + k, \beta + k, \dots$

Let $y = x + k$, so $x = y - k$. Substitute into the original polynomial.

Given α, β roots of $x^2 - 5x + 6 = 0$, find quadratic with roots $\alpha + 2, \beta + 2$.
 Sub $y = x + 2$, $x = y - 2$: $(y - 2)^2 - 5(y - 2) + 6 = 0$. Expand: $y^2 - 9y + 20 = 0$.

Reciprocal: roots $1/\alpha, 1/\beta, \dots$

Sub $y = 1/x$ then $x = 1/y$. For polynomial coefficients $a_n, \dots, a_0$, this reverses the coefficients.

Cubic with roots reciprocals of $x^3 - 3x^2 + 2x - 1 = 0$.
 Reverse coefficients: $-y^3 + 2y^2 - 3y + 1 = 0$, or $y^3 - 2y^2 + 3y - 1 = 0$.

Scale: roots $k\alpha, k\beta, \dots$

Sub $y = kx$, $x = y/k$.

4. Worked Examples

Given roots α, β, γ of $x^3 - 6x^2 + 11x - 6 = 0$, find $\sum \alpha^2$.

$\sum \alpha = 6, \sum \alpha\beta = 11, \alpha\beta\gamma = 6. \sum \alpha^2 = 36 - 22 = 14.$

Given α, β roots of $x^2 - 7x + 2 = 0$, form quadratic with roots α^2, β^2 .

$\alpha^2 + \beta^2 = 49 - 4 = 45; (\alpha\beta)^2 = 4.$ Quadratic: $y^2 - 45y + 4 = 0.$

5. Higher-Order Roots (Newton's Identities)

Let $S_n = \alpha^n + \beta^n + \dots$. For quadratic: $S_1 = \alpha + \beta, S_2 = (\alpha + \beta)^2 - 2\alpha\beta$, and recursively $S_n = (\alpha + \beta)S_{n-1} - \alpha\beta S_{n-2}$.

For $x^2 - 3x + 1 = 0$, find $\alpha^5 + \beta^5$.

$S_0 = 2, S_1 = 3, S_2 = 9 - 2 = 7, S_3 = 3(7) - 1(3) = 18, S_4 = 3(18) - 7 = 47, S_5 = 3(47) - 18 = 123.$

6. Exam Strategy

- Memorise the sign pattern and Vieta's formulae for quadratic, cubic, quartic.
- Symmetric-function identities: derive them once; re-use.
- Transformations via substitution are usually safer than re-deriving sum/product of new roots.
- Give exact answers; don't approximate.